1 Flow equation example

$$\partial\Gamma_k[\Phi] = \tag{1}$$

$$= \left[k^2\Theta(k^2-p^2) \left(\frac{(-\partial_\mu^2 + 4U^{'} + [4\phi_{1x}^2 + 4\phi_{2x}^2]U^{''})\delta^{(d)}(x-y) + 2(k^2-p^2)\Theta(k^2-p^2)}{-16U^{''}\phi_{1x}\phi_{1y}\phi_{2x}\phi_{2y} + \Delta_1\Delta_2} \right) \right] \tag{2}$$

where

$$\Delta_1 := \left((-\frac{1}{2}\partial_\mu^2 + 2U^{'} + 4U^{''}\phi_{1x}^2)\delta^{(d)}(x-y) + \Theta(k^2-p^2) \right) \tag{3}$$

$$\Delta_2 := \left((-\frac{1}{2}\partial_\mu^2 + 2U^{'} + 4U^{''}\phi_{2x}^2)\delta^{(d)}(x-y) + \Theta(k^2-p^2) \right) \tag{4}$$

$$U := U(\phi) \tag{5}$$

$$\phi_{1x} := \phi_1(x) \tag{6}$$

$$\phi_{2x} := \phi_2(x) \tag{7}$$

$$\phi_{1y} := \phi_1(y) \tag{8}$$

$$\phi_{2x} := \phi_2(y) \tag{9}$$

$$\tag{10}$$